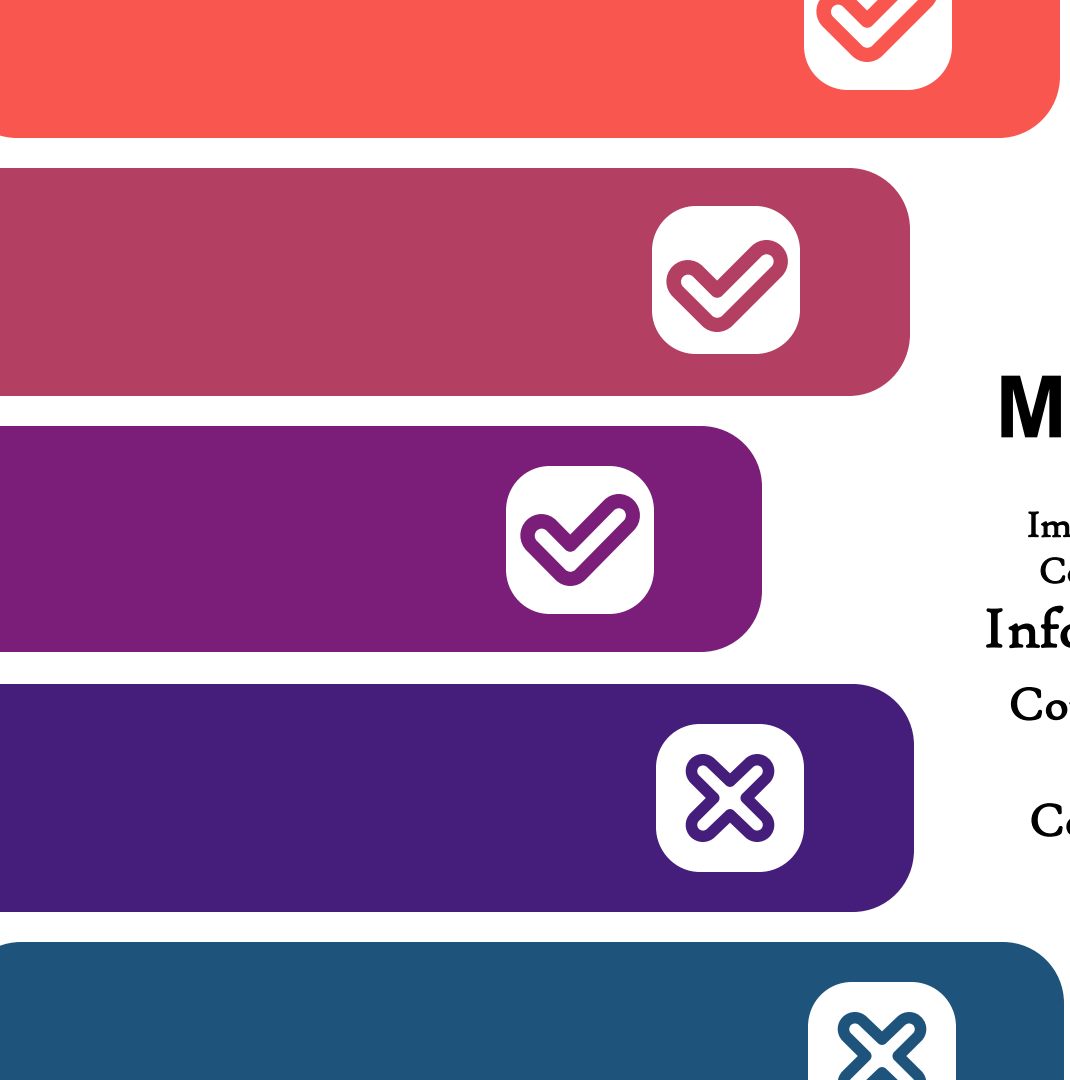


Intelligent Task Management System

Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences
Information Systems Department

Course Title : Introduction to Databases
Course Code : IS 1220
Course Instructor : Ms. Nada Alogaiel

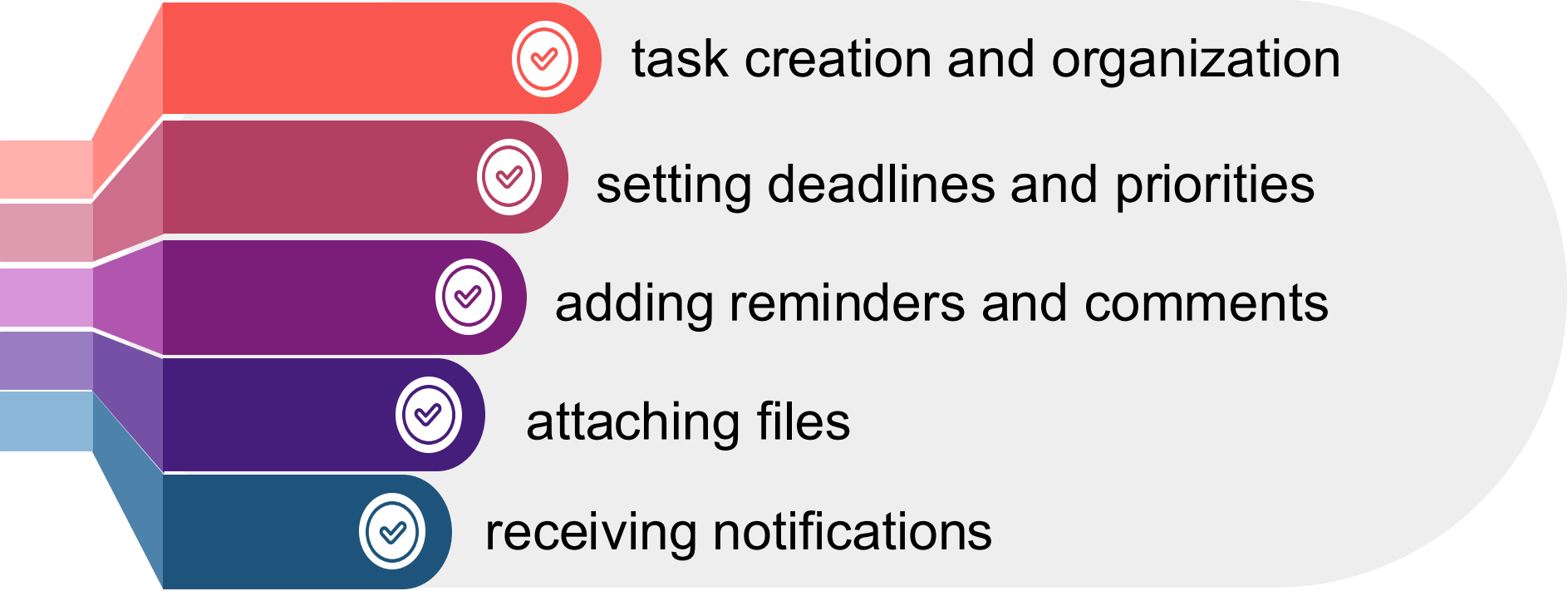
Students Names :
Norah Alkharan 445000782
Reema Alshahrani 445002257
Sara Alhulaisy 445016161

Agenda

- Project Description
- Functional Requirements
- EER Diagram Overview
- Mapping
- Normalization
- SQL Implementation
- Sample Queries
- User Interface Overview

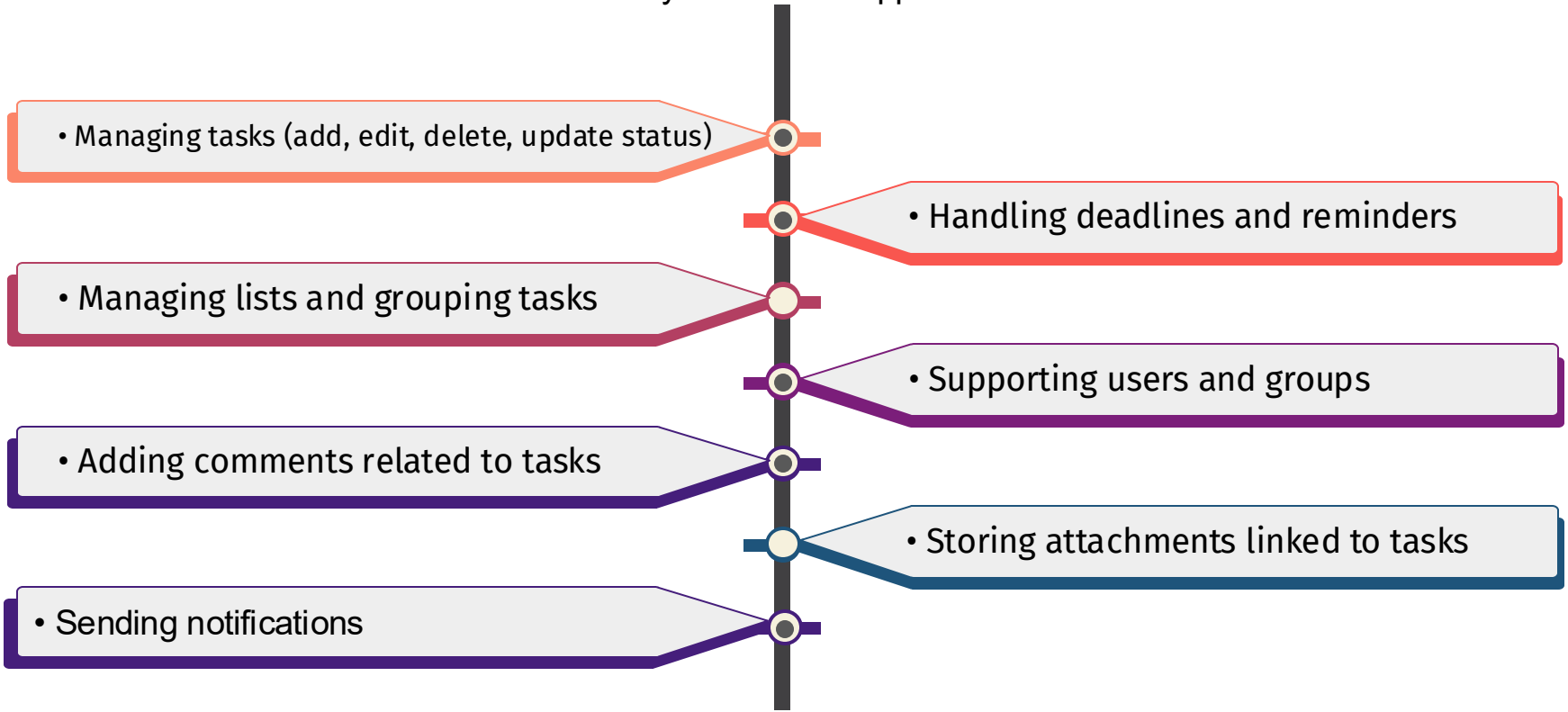
Project Description

The Intelligent Task Management System helps users manage both individual and group tasks efficiently.



Functional Requirements

The system must support:

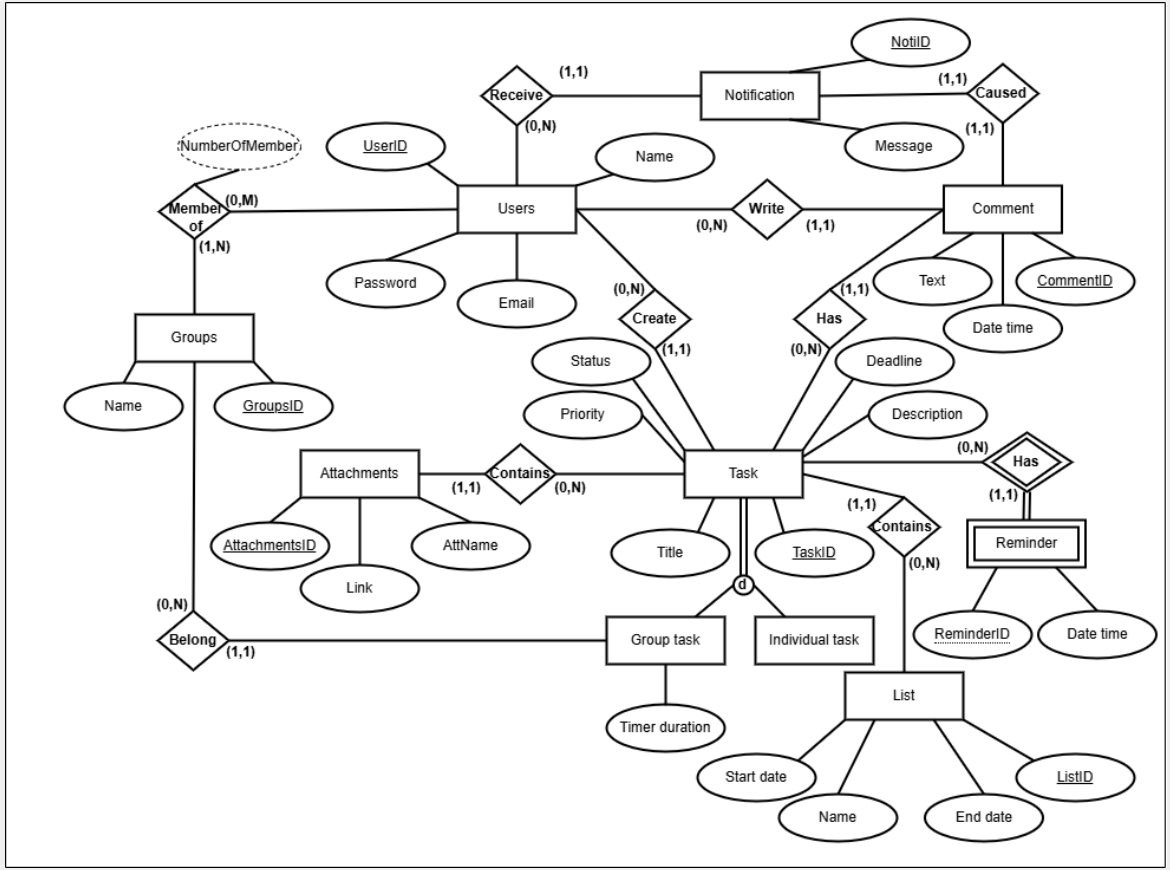


EER Diagram Overview

The conceptual model contains:

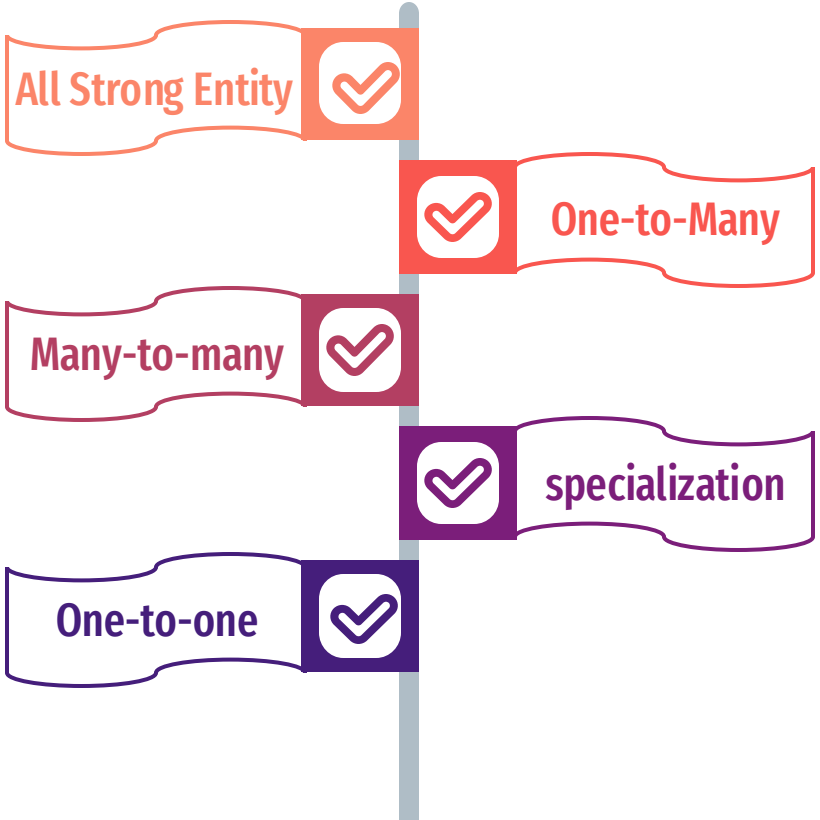
- User, Task, List, Groups, Comment, Reminder, Attachment, Notification
- Task specialization into Group Task and Individual Task
- Users can join multiple groups
- Group tasks belong to a single group
- Comments generate notifications
- Tasks may have reminders and attachments

EER Diagram


NotiID
Receive
(1,1)
Notification
(1,1)
Caused
(0,N)
Message
(1,1)
NumberOfMember
UserID
Name
Member of
(0,M)
(1,N)
Users
Write
(0,N)
(1,1)
Comment
Password
Email
(0,N)
Create
(1,1)
(1,1)
Has
(0,N)
Text
CommentID
Date time
Groups
Name
GroupsID
Status
Priority
Deadline
Description
(0,N)
Has
(1,1)
Attachments
(1,1)
Contains
(0,N)
Task
(1,1)
Contains
(0,N)
Reminder
AttachmentsID
Link
AttName
Title
TaskID
d
(0,N)
Belong
(1,1)
Group task
Individual task
ReminderID
Date time
List
Timer duration
Start date
Name
End date
ListID

Mapping Summary

- Each entity was converted into a separate table with its own primary key



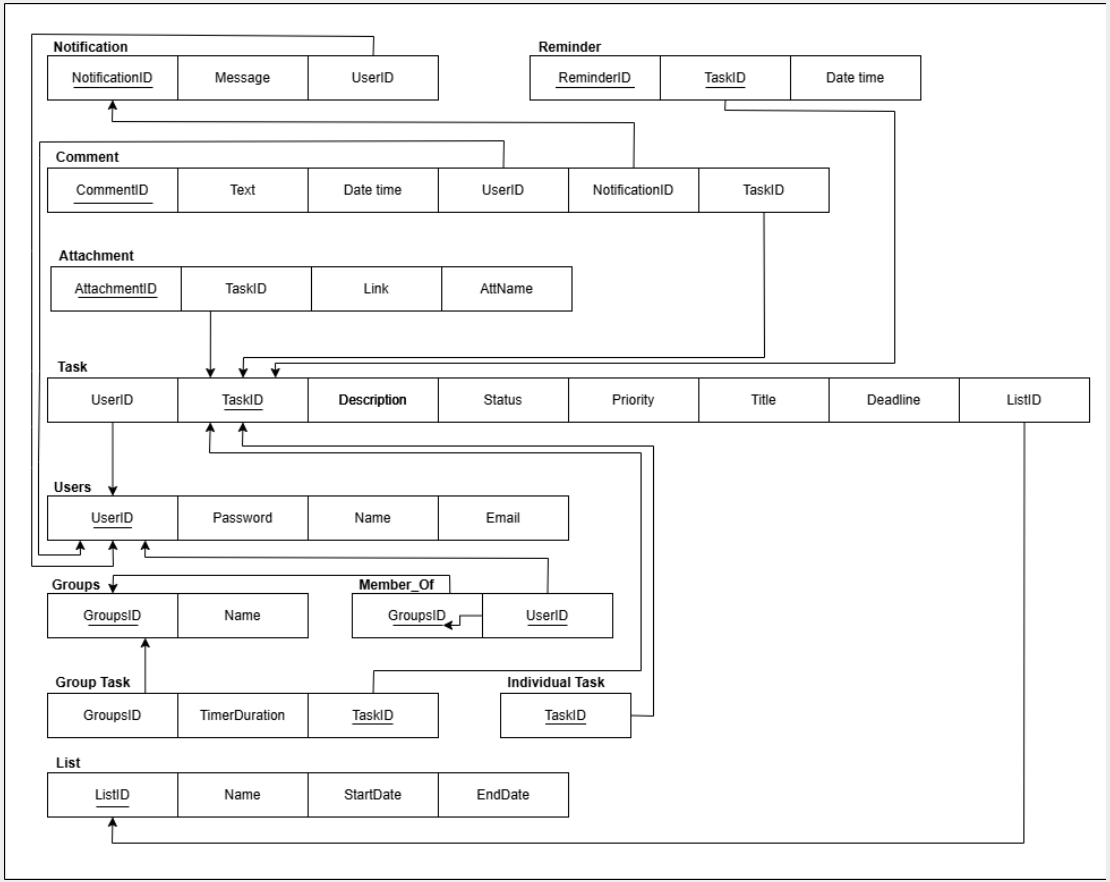
- In every One-to-Many relationship, the foreign key is placed on the “many” side

- Many-to-many between Users and Groups handled through Member_Of

- Task specialization implemented using 8A with two subclasses tables

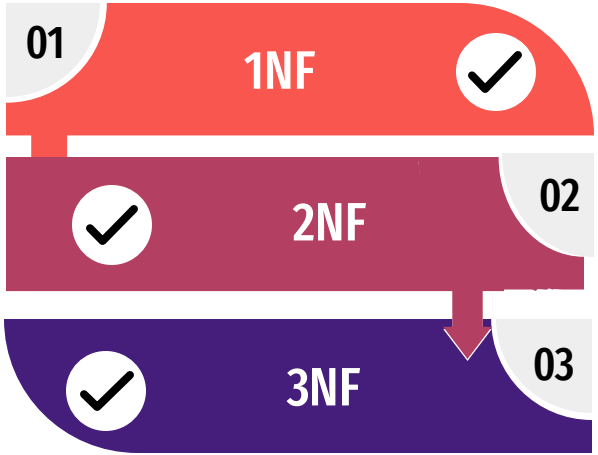
- One-to-one between Comment and Notification mapped using NotificationID in Comment

Mapping


Notification
NotificationID
Message
UserID
Reminder
ReminderID
TaskID
Date time
Comment
CommentID
Text
Date time
UserID
NotificationID
TaskID
Attachment
AttachmentID
TaskID
Link
AttName
Task
UserID
TaskID
Description
Status
Priority
Title
Deadline
ListID
Users
UserID
Password
Name
Email
Groups
GroupsID
Name
Member_Of
GroupsID
UserID
Group Task
GroupsID
TimerDuration
TaskID
Individual Task
TaskID
List
ListID
Name
StartDate
EndDate

Normalization

- No repeated groups
- No MV Attributes



- No partial dependency
- Only Member_Of uses composite key and attributes depend on full key

- No transitive dependency
- Every non-key field depends directly on its table's primary key

Create Database

```
CREATE TABLE Users (
    UserID          CHAR(10)        NOT NULL,
    UsersPassword   VARCHAR(100)    NOT NULL,
    UsersName       VARCHAR(100)    NOT NULL,
    Email           VARCHAR(150)    NOT NULL,
    PRIMARY KEY (UserID)
);

CREATE TABLE List (
    ListID          INT             NOT NULL AUTO_INCREMENT,
    ListName        VARCHAR(100)    NOT NULL,
    StartDate       DATE            NULL,
    EndDate         DATE            NULL,
    PRIMARY KEY (ListID)
);
CREATE TABLE GGroups (
    GroupID         INT             NOT NULL AUTO_INCREMENT,
    GroupsName      VARCHAR(100)    NOT NULL,
    PRIMARY KEY (GroupsID)
);
```

Create Database

```
CREATE TABLE Task (
    TaskID          INT             NOT NULL AUTO_INCREMENT,
    UserID          CHAR(10)        NOT NULL,
    Description     VARCHAR(500)    NULL,
    Status          VARCHAR(20)     NULL,
    Priority        VARCHAR(20)     NULL,
    Title           VARCHAR(100)    NOT NULL,
    Deadline        DATETIME        NULL,
    ListID          INT             NULL,
    PRIMARY KEY (TaskID),
    FOREIGN KEY (UserID) REFERENCES Users(UserID) ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (ListID) REFERENCES List(ListID) ON DELETE SET NULL ON UPDATE CASCADE
);
CREATE TABLE Notification (
    NotificationID  INT             NOT NULL AUTO_INCREMENT,
    Message         TEXT            NOT NULL,
    UserID          CHAR(10)        NOT NULL,
    PRIMARY KEY (NotificationID),
    FOREIGN KEY (UserID) REFERENCES Users(UserID) ON DELETE CASCADE ON UPDATE CASCADE
);
```

Create Database

```
CREATE TABLE Reminder (
    ReminderID       INT                NOT NULL AUTO_INCREMENT,
    TaskID           INT                NOT NULL,
    ReminderDateTime DATETIME           NOT NULL,
    PRIMARY KEY (ReminderID),
    FOREIGN KEY (TaskID) REFERENCES Task(TaskID) ON DELETE CASCADE ON UPDATE CASCADE
);
CREATE TABLE Comment (
    CommentID        INT                NOT NULL AUTO_INCREMENT,
    CommentText      TEXT               NOT NULL,
    CommentDateTime  DATETIME           NOT NULL,
    UserID           CHAR(10)           NOT NULL,
    NotificationID   INT                NULL,
    TaskID           INT                NULL,
    PRIMARY KEY (CommentID),
    FOREIGN KEY (UserID) REFERENCES Users(UserID) ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (NotificationID) REFERENCES Notification(NotificationID) ON DELETE SET NULL ON UPDATE CASCADE,
    FOREIGN KEY (TaskID) REFERENCES Task(TaskID) ON DELETE SET NULL ON UPDATE CASCADE
);
```



Create Database



```
CREATE TABLE Member_Of (
    GroupsID        INT             NOT NULL,
    UserID          CHAR(10)        NOT NULL,
    PRIMARY KEY (GroupsID, UserID),
    FOREIGN KEY (GroupsID) REFERENCES GGroups(GroupsID) ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (UserID) REFERENCES Users(UserID) ON DELETE CASCADE ON UPDATE CASCADE
);
CREATE TABLE GroupTask (
    GroupsID        INT             NOT NULL,
    TaskID          INT             NOT NULL,
    TimerDuration   INT             NULL,
    PRIMARY KEY (GroupsID, TaskID),
    FOREIGN KEY (GroupsID) REFERENCES GGroups(GroupsID) ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (TaskID) REFERENCES Task(TaskID) ON DELETE CASCADE ON UPDATE CASCADE
);
```

Create Database

```
CREATE TABLE Attachment (
    AttachmentID     INT                NOT NULL AUTO_INCREMENT,
    TaskID           INT                NOT NULL,
    Link             VARCHAR(500)       NOT NULL,
    AttName          VARCHAR(100)       NOT NULL,
    PRIMARY KEY (AttachmentID),
    FOREIGN KEY (TaskID) REFERENCES Task(TaskID) ON DELETE CASCADE ON UPDATE CASCADE
);
CREATE TABLE IndividualTask (
    TaskID           INT                NOT NULL,
    PRIMARY KEY (TaskID),
    FOREIGN KEY (TaskID) REFERENCES Task(TaskID) ON DELETE CASCADE ON UPDATE CASCADE
);
```

Insert values

```
INSERT INTO Users (UserID, UsersPassword, UsersName, Email) VALUES
('U000000001', 'Reema$123',       'Reema Ali',  'reemaaali.1426@gmail.com'),
('U000000002', 'Sara2005',        'Sara',       'sara.2005@gmail.com'),
('U000000003', 'Nourah_saleh',    'Nourah',     'Nourah.1357@gmail.com'),
('U000000004', 'eng447',          'Khalid',     'Eng.Khalid@gmail.com'),
('U000000005', 'nora.f',          'Nora Fahad', 'nora.fahad@gmail.com'),
('U000000006', 'Aishah.1350',     'Aishah',     'Aishah.ahmmad@gmail.com'),
('U000000007', 'Gh.ali1418',      'Ghadah',     'Ghadah.ali.1418@gmail.com');
```

Insert values

```
INSERT INTO List (ListName, StartDate, EndDate) VALUES
('Study plan',       '2025-12-01', '2026-01-08'),
('Workshop',         '2025-02-01', '2025-02-28'),
('Exam Schedule',    '2025-12-28', '2026-01-08'),
('Designing list',    NULL,         NULL),
('Shopping plan',    '2026-01-01', '2026-01-30'),
('Self-Improvement', '2026-01-09', '2026-02-09'),
('Healthy food',     '2026-01-01',  NULL);
```

Insert values

```
INSERT INTO GGroups (GroupsName) VALUES
('Project Data Base Team'),
('Community Team'),
('Council Team'),
('IT Group'),
('Workshops Team'),
('Data analysis Team'),
('Cybersecurity Team');
```

Insert values

```
INSERT INTO Task (UserID, Description, Status, Priority, Title, Deadline, ListID) VALUES
('U000000001', 'Finish the DBMS project',   'In Progress', 'High',  'DBMS Project',         '2025-12-7 10:00:00',  1),
('U000000002', 'Write the documentation',   'In Progress', 'High',  'Document',             '2025-12-5 14:00:00',  1),
('U000000003', 'Prepare the presentation',  'Pending',     'High',  'Presentation',         '2025-12-8 09:00:00',  1),
('U000000004', 'Complete the AI course',    'Completed',   'Low',   'AI',                    NULL,                 4),
('U000000005', 'Study SQL',                 'Pending',     'High',  'SQL Study',            '2025-12-6 17:00:00',  5),
('U000000006', 'Complete the course',       'Pending',     'Low',   'Personal development', '2026-02-09 10:00:00', 6),
('U000000007', 'Join the club',             'Pending',     'Low',   'Workout Plan',          NULL,                 7);
```

Insert values

```
INSERT INTO Notification (Message, UserID) VALUES
('Don’t miss the completed task', 'U000000001'),
('You have a new group!',         'U000000002'),
('You have a new comment',        'U000000003'),
('Task completed!',               'U000000004'),
('Keep going',                    'U000000005'),
('You almost at the end!',        'U000000006'),
('Do not forget',                 'U000000007');
```

Insert values

```
INSERT INTO Reminder (TaskID, ReminderDateTime) VALUES
(1, '2025-12-04 09:00:00'),
(2, '2025-12-05 12:00:00'),
(3, '2025-12-06 08:00:00'),
(4, '2025-12-07 15:00:00'),
(5, '2025-12-18 16:00:00'),
(6, '2026-02-09 09:00:00'),
(7, '2026-03-03 12:00:00');
```

Insert values

```
INSERT INTO Comment (CommentText, CommentDateTime, UserID, NotificationID, TaskID) VALUES
('Good luck!',                    '2025-01-02 10:00:00', 'U000000001', 1, 1),
('Do you want to add a new user?', '2025-01-05 12:30:00', 'U000000002', 2, 2),
('Presentation looks good.',      '2025-01-06 14:00:00', 'U000000003', 3, 3),
('Well done',                     '2025-01-07 09:15:00', 'U000000004', 2, 4),
('Studying SQL now',              '2025-01-08 18:00:00', 'U000000005', 5, 5),
('Great progress!',               '2026-02-06 11:00:00', 'U000000006', 6, 6),
('Keep going!',                   '2026-03-04 13:00:00', 'U000000007', 2, 7);
```

Insert values

```
INSERT INTO Member_Of (GroupsID, UserID) VALUES
(1, 'U000000001'),
(1, 'U000000002'),
(2, 'U000000003'),
(3, 'U000000004'),
(1, 'U000000005'),
(3, 'U000000006'),
(3, 'U000000007');
```

Insert values

```
INSERT INTO GroupTask (GroupsID, TaskID, TimerDuration) VALUES
(1, 1, 30),
(1, 2, 45),
(2, 3, 20),
(3, 4, 60),
(4, 5, 25),
(6, 6, 40),
(7, 7, 55);
```

Insert values

```
INSERT INTO Attachment (TaskID, Link, AttName) VALUES
(1, 'https://tinyurl.com/4rj6phu7',                                        'ProjectDBMS.pdf'),
(2, 'https://tinyurl.com/jctnbtd2',                                        'documentation.docx'),
(3, 'https://tinyurl.com/jctnbtd2',                                        'presentation.pptx'),
(4, 'https://zc1.maillist-manage.in/click/11feeefb0cb80a5c/11feeefb0ca388dc', 'list.jpg'),
(5, 'https://youtu.be/-_wLgK0IKfo?si=hM7xWUd5cdAreIfq',                    'sql.mov'),
(5, 'https://satr.tuwaiq.edu.sa',                                          'Satr Course Link'),
(4, 'https://samai.futurex.sa',                                            'Samai Course Link');
```

Insert values

```
INSERT INTO IndividualTask (TaskID) VALUES
(1),
(2),
(3),
(4),
(5),
(6),
(7);
```

User table

	UserID	UsersPassword	UsersName	Email
▶	U000000001	Reema$123	Reema Ali	reemaaali.1426@gmail.com
	U000000002	Sara2005	Sara	sara.2005@gmail.com
	U000000003	Nourah_saleh	Nourah	Nourah.1357@gmail.com
	U000000004	eng447	Khalid	Eng.Khalid@gmail.com
	U000000005	nora.f	Nora Fahad	nora.fahad@gmail.com
	U000000006	Aishah.1350	Aishah	Aishah.ahmmad@gmail.com
	U000000007	Gh.ali1418	Ghadah	Ghadah.ali.1418@gmail.com
*	NULL	NULL	NULL	NULL

List table

ListID	ListName	StartDate	EndDate
1	Study plan	2025-12-01	2026-01-08
2	Workshop	2025-02-01	2025-02-28
3	Exam Schedule	2025-12-28	2026-01-08
4	Designing list	NULL	NULL
5	Shopping plan	2026-01-01	2026-01-30
6	Self-Improvement	2026-01-09	2026-02-09
7	Healthy food	2026-01-01	NULL
NULL	NULL	NULL	NULL

Group table

	GroupSID	GroupsName
▶	1	Project Data Base Team
	2	Community Team
	3	Council Team
	4	IT Group
	5	Workshops Team
	6	Data analysis Team
	7	Cybersecurity Team
*	NULL	NULL

Task table

TaskID	UserID	Description	Status	Priority	Title	Deadline	ListID
1	U000000001	Finish the DBMS project	In Progress	High	DBMS Project	2025-12-07 10:00:00	1
2	U000000002	Write the documentation	In Progress	High	Document	2025-12-05 14:00:00	1
3	U000000003	Prepare the presentation	Pending	High	Presentation	2025-12-08 09:00:00	1
4	U000000004	Complete the AI course	Completed	Low	AI	NULL	4
5	U000000005	Study SQL	Pending	High	SQL Study	2025-12-06 17:00:00	5
6	U000000006	Complete the course	Pending	Low	Personal development	2026-02-09 10:00:00	6
7	U000000007	Join the club	Pending	Low	Workout Plan	NULL	7
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Notification table

NotificationID	Message	UserID
1	Don't miss the completed task	U000000001
2	You have a new group!	U000000002
3	You have a new comment	U000000003
4	Task completed!	U000000004
5	Keep going	U000000005
6	You almost at the end!	U000000006
7	Do not forget	U000000007
NULL	NULL	NULL

Reminder table

	ReminderID	TaskID	ReminderDateTime
▶	1	1	2025-12-04 09:00:00
	2	2	2025-12-05 12:00:00
	3	3	2025-12-06 08:00:00
	4	4	2025-12-07 15:00:00
	5	5	2025-12-18 16:00:00
	6	6	2026-02-09 09:00:00
	7	7	2026-03-03 12:00:00
*	NULL	NULL	NULL

Comment table

CommentID	CommentText	CommentDateTime	UserID	NotificationID	TaskID
1	Good luck!	2025-01-02 10:00:00	U000000001	1	1
2	Do you want to add a new user?	2025-01-05 12:30:00	U000000002	2	2
3	Presentation looks good.	2025-01-06 14:00:00	U000000003	3	3
4	Well done	2025-01-07 09:15:00	U000000004	2	4
5	Studying SQL now	2025-01-08 18:00:00	U000000005	5	5
6	Great progress!	2026-02-06 11:00:00	U000000006	6	6
7	Keep going!	2026-03-04 13:00:00	U000000007	2	7
NULL	NULL	NULL	NULL	NULL	NULL

Member_of table

	GroupsID	UserID
▶	1	U000000001
	1	U000000002
	2	U000000003
	3	U000000004
	1	U000000005
	3	U000000006
	3	U000000007
*	NULL	NULL

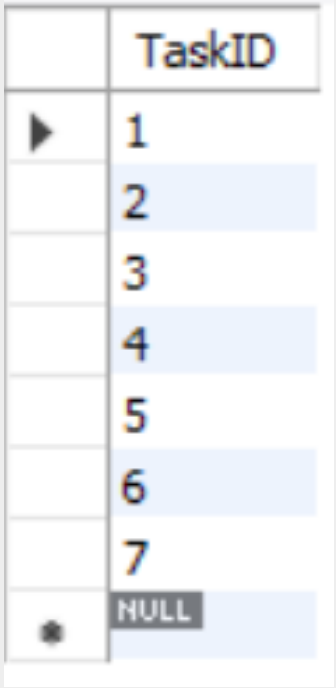
Group task table

	GroupsID	TaskID	TimerDuration
▶	1	1	30
	1	2	45
	2	3	20
	3	4	60
	4	5	25
	6	6	40
	7	7	55
*	NULL	NULL	NULL

Attachment table

AttachmentID	TaskID	Link	AttName
1	1	https://tinyurl.com/4rj6phu7	ProjectDBMS.pdf
2	2	https://tinyurl.com/jctnbtd2	documentation.docx
3	3	https://tinyurl.com/jctnbtd2	presentation.pptx
4	4	https://zc1.maillist-manage.in/click/11feeefb0cb...	list.jpg
5	5	https://youtu.be/-_wLgK0IKfo?si=hM7xWUd5c...	sql.mov
6	5	https://satr.tuwaiq.edu.sa	Satr Course Link
7	4	https://samai.futurex.sa	Samai Course Link
NULL	NULL	NULL	NULL

Individual table



	TaskID
▶	1
	2
	3
	4
	5
	6
	7
*	NULL

Update Queries

```sql
UPDATE Users
SET UsersName = 'Eng.Khalid'
WHERE UserID = 'U000000004';
```

After update

```sql
SELECT  UserID, UsersName
FROM Users
WHERE UserID = 'U000000004';
```

Result

	UserID	UsersName
▶	U000000004	Eng.Khalid
*	NULL	NULL

Update Queries

```
UPDATE Task
SET Priority = 'Medium'
WHERE UserID = 'U000000003';
```

After update

```
SELECT   UserID, Priority
FROM     Task
WHERE    UserID = 'U000000003';
```

Result

	UserID	Priority
▶	U000000003	Medium

Delete Queries

```
DELETE FROM Notification
WHERE       NotificationID = '4' ;
```

After delete

```
SELECT   Message, UserID
FROM     Notification
WHERE    NotificationID = '4';
```

Result

	Message	UserID

Nested Queries

```
SELECT Title, UserID
FROM Task
WHERE UserID IN (
    SELECT UserID
    FROM Notification
);
```

Result

	Title	UserID
▶	DBMS Project	U000000001
	Document	U000000002
	Presentation	U000000003
	SQL Study	U000000005
	Personal development	U000000006
	Workout Plan	U000000007

Nested Queries

```
WHERE UserID IN (
    SELECT UserID
    FROM Task
    WHERE TaskID IN (SELECT TaskID FROM Attachment)
);
```

Result

UsersName
Reema Ali
Sara
Nourah
Khalid
Nora Fahad

Selsect Queries

```
SELECT *
FROM   Task
WHERE  priority = 'High';
```

	TaskID	UserID	Description	Status	Priority	Title	Deadline	ListID
▶	1	U000000001	Finish the DBMS project	In Progress	High	DBMS Project	2025-12-07 10:00:00	1
	2	U000000002	Write the documentation	In Progress	High	Document	2025-12-05 14:00:00	1
	5	U000000005	Study SQL	Pending	High	SQL Study	2025-12-06 17:00:00	5
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Selsect Queries

```
SELECT *
FROM     List
WHERE    StartDate BETWEEN '2025-1-01' AND '2025-12-31';
```

	ListID	ListName	StartDate	EndDate
▶	1	Study plan	2025-12-01	2026-01-08
	2	Workshop	2025-02-01	2025-02-28
	3	Exam Schedule	2025-12-28	2026-01-08
*	NULL	NULL	NULL	NULL

Join Queries

```
SELECT Task.Title, ListName
FROM Task
JOIN List ON Task.ListID = List.ListID;
```

	Title	ListName
▶	DBMS Project	Study plan
	Document	Study plan
	Presentation	Study plan
	AI	Designing list
	SQL Study	Shopping plan
	Personal development	Self-Improvement
	Workout Plan	Healthy food

Join Queries

```
SELECT UsersName, Task.Title
FROM Users
JOIN Task ON Users.UserID = Task.UserID;
```

	UsersName	Title
▶	Reema Ali	DBMS Project
	Sara	Document
	Nourah	Presentation
	Khalid	AI
	Nora Fahad	SQL Study
	Aishah	Personal development
	Ghadah	Workout Plan

Retrieval Queries

```
SELECT ListID, COUNT(*) AS TotalTasks
FROM Task
GROUP BY ListID;
```

	ListID	TotalTasks
▶	1	3
	4	1
	5	1
	6	1
	7	1

Retrieval Queries

```
SELECT TaskID, COUNT(*) AS AttachmentCount
FROM Attachment
GROUP BY TaskID
HAVING COUNT(*) > 1;
```

	TaskID	AttachmentCount
▶	4	2
	5	2

View

```
CREATE VIEW TasksAndStatus AS
SELECT Title, Status
FROM Task;
```

```
SELECT * FROM TasksAndStatus;
```

	Title	Status
▶	DBMS Project	In Progress
	Document	In Progress
	Presentation	Pending
	AI	Completed
	SQL Study	Pending
	Personal development	Pending
	Workout Plan	Pending

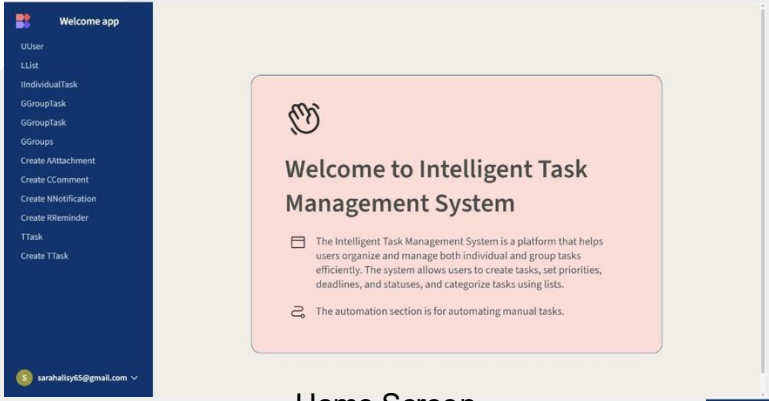
View

```sql
CREATE VIEW PendingTasks AS
SELECT TaskID, Title, Status
FROM Task
WHERE Status = 'Pending';
```

```sql
SELECT * FROM PendingTasks;
```

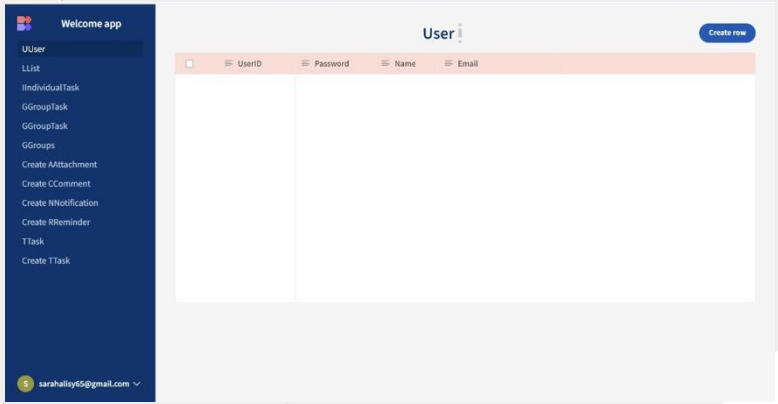
TaskID	Title	Status
3	Presentation	Pending
5	SQL Study	Pending
6	Personal development	Pending
7	Workout Plan	Pending

User Interface Screens



Home Screen

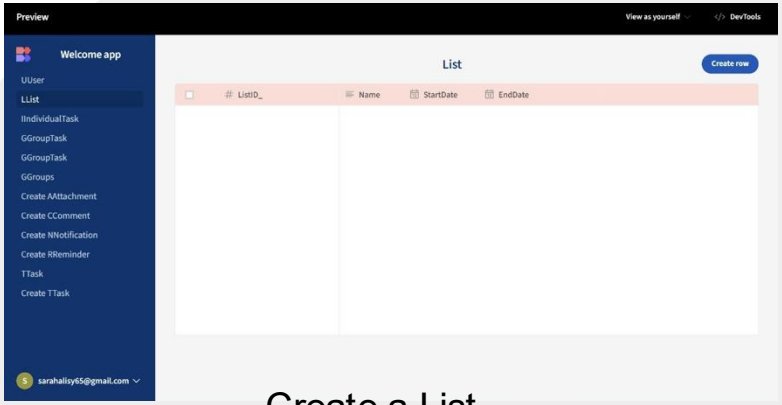
Provides the user with an overview of the site.



Create a User Account

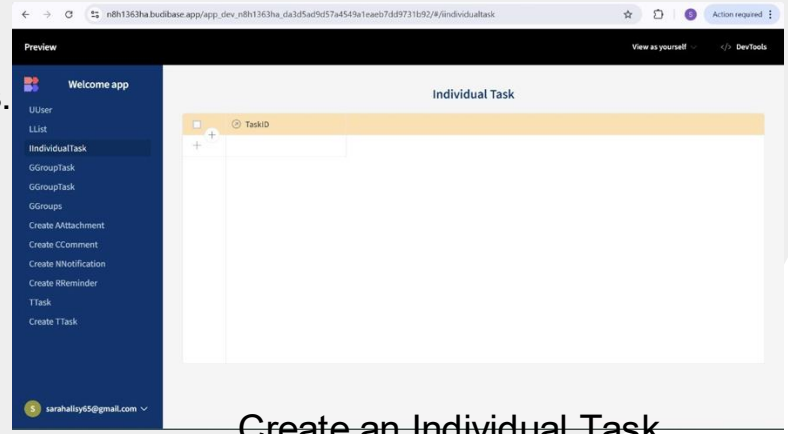
Allows the user to create a new account and enter their basic information.

User Interface Screens



Create a List

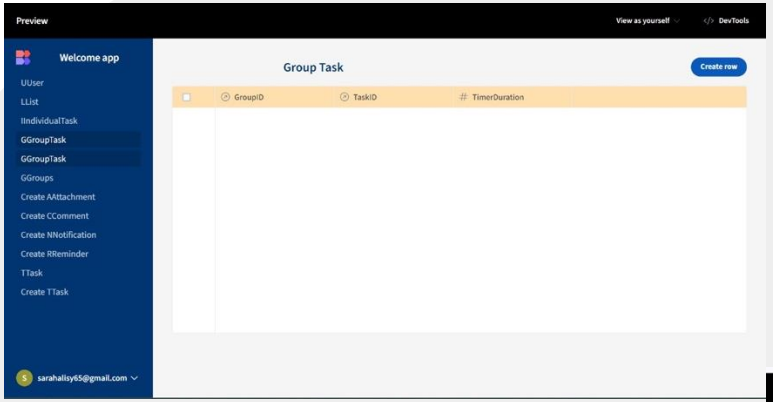
Enables the user to create a new list to organize tasks.



Create an Individual Task

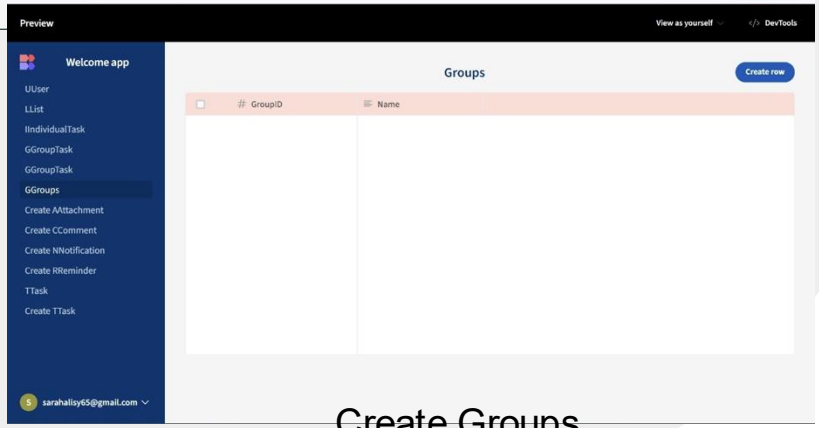
Page for creating a single task and specifying its details.

User Interface Screens



Create a Group Task

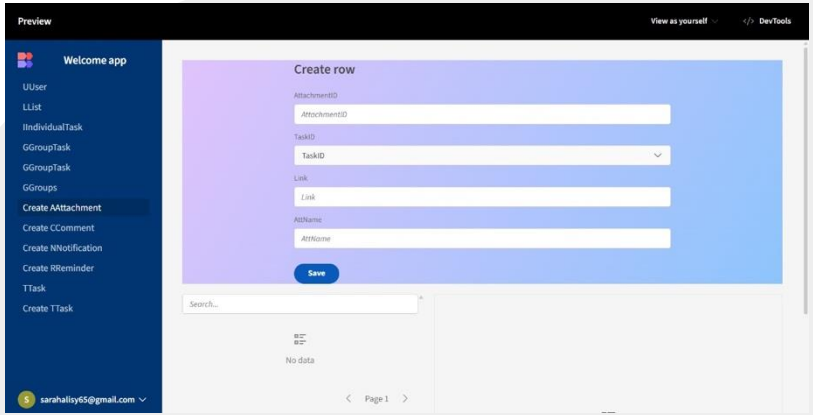
Allows the user to create a shared task and link it to a specific group.



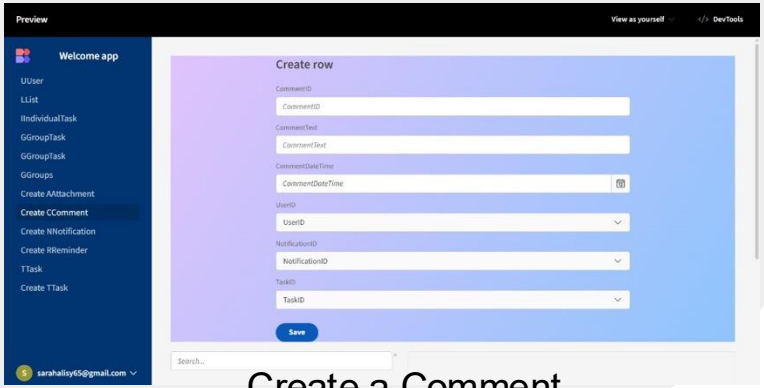
Create Groups

Page to create a new group and add members to it.

User Interface Screens

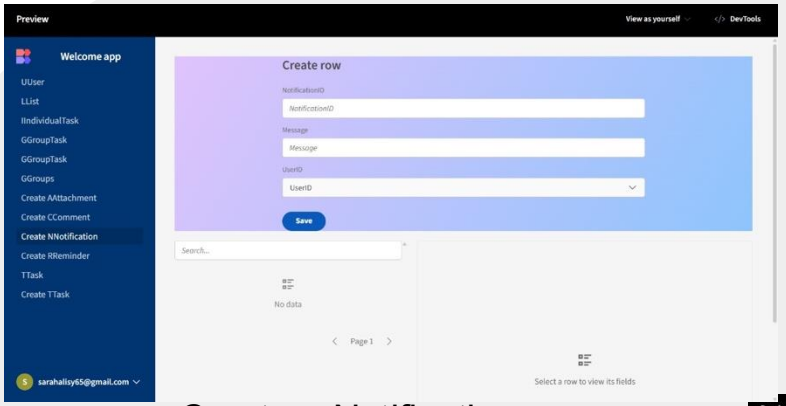


Create an Attachment
Allows the user to upload files or images and attach them to a task.



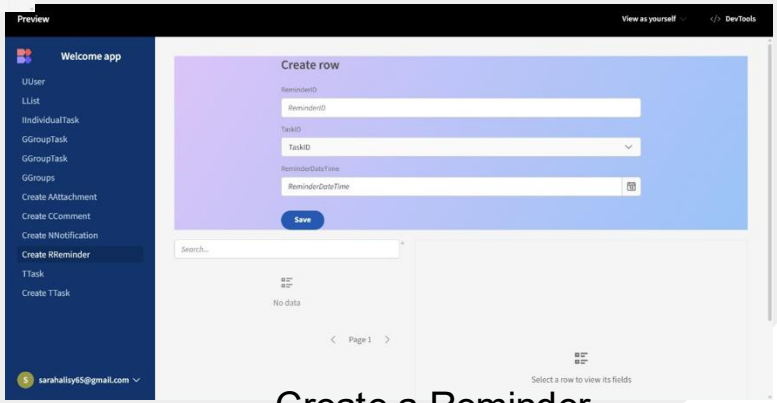
Create a Comment
Page to add a new comment within a task.

User Interface Screens



Create a Notification

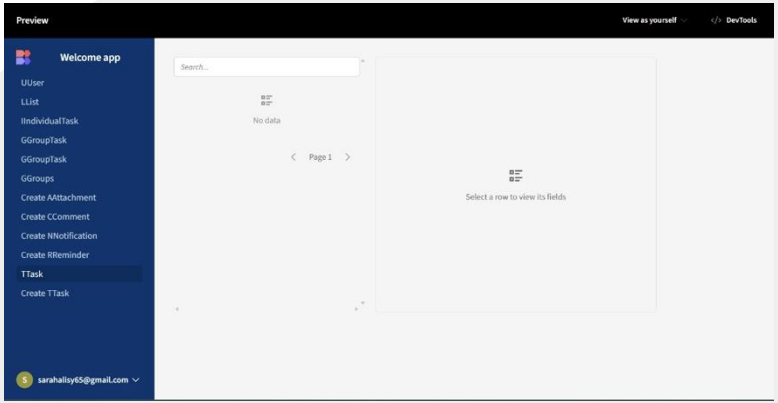
Generates a new notification to alert the user about a specific task.



Create a Reminder

Allows the user to create a reminder with a specified date for a task.

User Interface Screens



View Tasks

Displays all tasks to the user with options for sorting and searching.



Create a Task

A comprehensive page for creating a new task and specifying its details.

Conclusion

This project was a valuable learning experience.
We improved our understanding of database design and SQL implementation.
It strengthened our teamwork, problem-solving, and ability to turn requirements into a working system.

Thank you for listening

Presented By:
Norah Alkharan 445000782
Reema Alshahrani 445002257
Sara Alhulaisy 445016161